



*Leaders in Mobile Programming*



*Ingenious Problem Solving  
Swift Product Development  
Novel Value-Added Solutions*

*"...all the little choices they made were always perfect, giving our application a professional and nicely finished touch"*  
Patrice Uzan - VP , Natural Widgets

*"Degel has helped us with key pieces..., doing top quality work... these guys are true professionals... a pleasure to work with"*  
Nancy Yavne GM,  
Sun Israel Dev. Center

# *Value-Added Client Applications*

Degel Software Ltd. provides technical solutions in the mobile applications market. We address the full development cycle from pre-project planning to post-project follow-through.

We offer ingenious problem solving and project management, ensuring that projects move forward quickly and successfully. We create novel value-added solutions for mobile devices:

- productivity applets
- client-server, server-mediated, and P2P applications
- porting of desktop applications to mobile platforms or porting between mobile platforms

Whether you need J2ME, Symbian OS, BREW, Palm OS, Blackberry, or Pocket PC, there are no limits to what you can accomplish with Degel.

We can work with your development teams to optimize final products or with your marketing teams to turn “blue sky” ideas into exciting proofs-of-concept.



# Design Services

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## *Analysis of Customer Needs*

During the dot-com bubble, many companies believed that “if you build it, they will come.” Today, your marketing department is asking, “if you build it, will they care?” and your CTO is asking, “if we promise it, can we deliver it on time?” Degel can help your marketing team mesh customer expectations with the limitations of the technically feasible. We help you maximize the value of your R&D effort and get your development team in sync with your marketing requirements.

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## *Timetable and Roadmap*

Degel can help you build a technology roadmap.

- By explaining the benefits and consequences of new technologies, we help you estimate their impact on your schedule.
- Using our proven estimation methodologies, we help you schedule better and create accurate milestone estimates.

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## *Protocol Design*

Today's mobile applications require network connectivity. Degel has been designing network protocols for years and can create efficient protocols to meet your needs: UDP, TCP, HTTP, or WAP bearer; client-server, server-proxied, or peer-to-peer topology; Java or C++.

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## *GUI Design*

Mobile GUI design is challenging. Screen sizes are small, standard controls are limited, keypads are non-standard, and each manufacturer has different design rules. Whether your goal is to match the design of the target platform, emulate another device, or create your own branded look and feel, we will design the GUI you need.

# Development Services

## Deep Symbian Experience

Degel is proud to be a Symbian Platinum Partner. We have been programming Symbian OS since the Revo, continuing through to today's exciting Nokia Series 60, 80, and 90 as well as UIQ devices. We have built everything from Bluetooth to instant messaging to call recorders.



- If you want to port your Palm or Pocket PC app to Symbian... Degel can do it!
- If you want to do the undocumented on Symbian... Degel can do it!
- If you want to do the impossible on Symbian... Degel can do it! (Until we did it, everyone "knew" that you *cannot* add a device driver to the Nokia 9210).

## Cross-Target J2ME Development

Degel's cross-target tools allow us to cut through the jungle of J2ME vendor incompatibilities, with support for major vendors, conditional compilation, and localization.

- "One-touch build" eases development for multiple platforms and locales.
- Custom tools diagnose vendor problems with memory, GC, threading, networking, and SMS. Often, we find vendor bugs before the vendors do.

We have written UI libraries, productivity apps, and networked applications for J2ME. Our code runs on devices from Motorola, Nokia, Siemens, Sharp, Sony-Ericsson, and other vendors.

## Optimization Expertise

We are experts at profiling and optimization. We can determine your code's hotspots and devise techniques to accelerate performance and reduce memory consumption.

# Development Services

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## Expert Advice

Sometimes, your engineers just need an occasional hand or a nudge in the right direction. Many of our clients address this need with our *expert advisors* service, where we offer access to our team of experts and rapid answers to questions like:

- How can I access native handset features from my J2ME application?
- How can I create one application for Palm, Brew, and Symbian OS?
- How can we launch proprietary content from the Series 60 WAP viewer?
- Can we “sniff” IP traffic on a handset?

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## Components or Libraries

Often, our clients want to retain the bulk of the development process in-house. Degel can still assist with the creation of core components and libraries, supplementing your in-house expertise and supplying critical skills to meet your schedule.

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## Software Development

Degel's software gurus can undertake the full development of your projects including:

- design and coding
- project management and supervision
- code review and documentation
- software packaging and installation

# Development Services

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## **Prototypes and Demos**

If you need a prototype or demo, Degel is the answer. We specialize in rapid prototyping and are independent of your in-house scheduling pressures. We know when to implement each detail and when to just create a quick shell. We understand the importance of fast turnaround and rapid results. We know that the most important part of a successful prototype is leveraging existing tools and techniques.

If you need to create a fast demo or prototype, speak to us first!

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## **Internationalization and Localization**

Your applications must be prepared to enter the global market and address a variety of languages and cultures. Degel can assist you with software and the technical aspects of this task. We have completed many projects for the Japanese, American, European, and Israeli markets. Our code is internationalized and localization-ready. We can also guide you to resources that can assist with international marketing and localization.

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## **Project Management**

When you outsource a project to us, we can take responsibility for as much of the process as needed. With Degel you can expect:

- timely delivery
- detailed issues tracking
- smooth handover
- technical presentations, as needed, for your internal staff, investors, or customers.

You can rely on us for all the technical portions of your product delivery schedule.

# Coaching and Follow-Through

## *Coaching and Training*

Some of our clients like to do all of their core development in-house, but appreciate some assistance assimilating new technologies. For these clients, we offer coaching and training services to:

- find undocumented and under-documented platform APIs
- demonstrate and prototype correct API usage
- present effective platform tools and debugging techniques
- setup efficient mobile build environments
- assist in debugging tough problems
- enhance productivity on each mobile platform.

## *Training on Mobile Development Tools*

Mobile development is very different from PC or Unix/Linux development. Each mobile device has its own toolset and development chain. Skills acquired in desktop development, or even on other mobile devices, do not transfer easily to each new device, handset, or emulator environment.

We can teach your development staff how to develop and maintain mobile applications. Whether you need to learn about Symbian APIs, J2ME quirks, platform dependencies, development hacks, or any other facet of mobile software development, we can bring you up-to-speed, quickly and painlessly.

## *Authoring of White Papers, Tech Reports, and Presentations*

Sometimes you need not only impromptu training for your staff, but also more formal documentation, whether for in-house use or for presentation to customers, investors, or partners. Here too, Degel can assist. We can produce lucid reports and presentations explaining areas of interest. With our help, you can show off your best work.

## Clients

*Degel has vast experience in the wireless arena. Here is a partial list of Degel's accomplishments for our clients. For a more complete list, please visit our website, <http://www.degel.com>.*



Degel is developing Press-to-Talk and other exciting next-generation mobile applications for Orange's Communications Products Organization.



Degel ported portions of Quicksheet to Nokia's Series 60. Quicksheet is an MS-Excel compatible application included in Quickoffice's suite of office applications.



Degel helped MobiMate to port their #1 travel application, WorldMate Professional Edition, to the Sony-Ericsson P800 and P900. Award-winning WorldMate offers up-to-date online flight schedules, online currency converter, weather, international time information, and more. It is the perfect application for world travelers.



Degel helped NextCode Corporation bring their innovative barcode scanning technology to mobile platforms.



## Clients



Degel ported Electric Pocket's BugMe! from Palm OS to Symbian OS, steering the design process to combine the spirit of the Palm version with the rich multimedia features of Nokia Series 60 and the Sony-Ericsson P800/P900.

Our work won Handango's 2003 Champion Award for best Symbian Productivity Application, and is the subject of a Forum Nokia case study.



Degel ported iambic's award-winning Agendus Palm contact manager to Symbian OS UIQ. Agendus's loyal followers demand a variety of rich GUI features and accelerators, which Degel successfully ported to Symbian.

A Handango 2004 Champion Award finalist!



Degel created Natural Widgets' NaturalRecorder, an exciting new Series 60 utility for automatically recording phone conversations.



Degel wrote Symbian device drivers for VKB's virtual keyboard, winning acclaim for VKB and Siemens at CeBIT 2004.



Degel ported Cash-U's gaming SDK to Symbian. Earlier, we devised an implementation strategy to allow them to take advantage of J2ME over-the-air provisioning.

## Clients



Degel developed ICQ's instant messaging (IM) client for many handheld platforms.

Our Symbian-based clients are the most powerful IM clients ever created for mobile devices.

Our J2ME-based ICQ clients pushed J2ME to the limit and uncovered network and GUI bugs in the implementations from Siemens, Motorola, Nokia, and other vendors. We helped the vendors resolve these issues, visiting their development offices to assure proper resolution.

Degel is very proud of having helped to bring high-end instant messaging to the mobile world.



Degel developed a Nokia 9210 mouse driver for OTM. We succeeded... even though both Nokia and popular opinion declared that there was no way to write an add-in device driver for the 9210.



For Sun's Israel Research Center, Degel added profiling capabilities to the KVM in the J2ME reference implementation, identified key performance issues, and doubled graphics performance. We also ported J2ME to several mobile platforms and created a device-independent graphics kernel for MIDP.



For Intel's wireless group in Israel, Degel evaluated the benefits and costs of porting a large Windows application to a wireless platform.

## FAQ - Frequently Asked Questions

### *How do you know this business?*

- Degel is run by industry expert David Goldfarb. His successes include Internet protocol development at MIT in the early 1980's, artificial intelligence at Symbolics and ILA; multimedia applications for Emblaze and Q Multimedia; and distributed applications with peer-to-peer (P2P) protocols at 2AM Development.
- We were involved in the early Arpanet, the AI craze, the Japanese boom, the multimedia madness of the early 90's, the rebirth of the Internet, the dot-com bubble, mobile technologies, and the current post-bubble recovery.
- Degel maintains a network of connections across the high-tech world in Israel, Europe, Japan, and both coasts of the United States, drawing upon some of the finest talent in the world.
- We are champions at problem solving, solution finding, reframing, and extrapolation. We can identify solutions from a wide range of resources and disciplines.
- We believe in continuous education. Our staff spends hours each week in ongoing research into technologies and techniques.

### *But, you're not expert at our niche!*

That's certainly true. You are the experts at your business. At Degel, we are experts in the areas that contribute to your technical success. We know a bit about microprocessor architecture, but we don't tell Intel how to make their chips. We know something about language design, but we don't tell Sun what features to put into the next version of Java. However, we have helped both Intel and Sun solve particular problems.

## FAQ - Frequently Asked Questions

### ***Are you outsourcers or management consultants?***

A bit of both, actually. Our background is very strong on the technology side, but we have run companies, served as CTOs, and managed highly visible development efforts. We have seen many companies, both successful and unsuccessful, and have learned from every encounter. We excel at bridging the gap between technologist and humanist; between R&D and marketing. We can identify the issues that affect your company's technological success and help you resolve the problems that interfere with your growth.

### ***Can't you just code our project?***

Sure! If you give us your specification, we can dive right in and implement it for you. Outsourcing is a major part of the value we provide. We're happy to offer you just the services you need to get your projects to market as quickly as possible.

### ***Our development team is top-notch; we can do our own coding.***

That's great! But, Degel can still help you.

- We can supply libraries or modules that will plug into the rest of your coding effort.
- We can coach your team, teach advanced techniques, and write prototypes to bring your team up-to-speed on new technologies.

### ***You will have too much control; we will lose process ownership!***

Degel's methodologies avoid this problem:

- We design our code for easy hand over.
- We believe in project follow-through. We will offer as much training as you want so that you are 100% comfortable with the code and the underlying technologies.
- We use version tracking and project management tools that make it easy to transfer even works-in-progress to you at any time.

# Think Outside the Box

At Degel Software Ltd., we specialize in creative thinking and in finding easy answers to impossible questions. Have fun with these questions, and see how well you do at solving the impossible!

- What comes next in this series?  
O, T, T, F, F, S, S, \_\_, \_\_, \_\_, \_\_,...
- You have twelve coins. Eleven weigh the same; one doesn't. With only three weighings on a balance scale, identify the bad coin and whether it is heavier or lighter. \_\_\_\_\_
- What is the longest month? \_\_\_\_\_
- Solve SEND+MORE=MONEY, where each letter represents a digit. \_\_\_\_\_
- Connect these nine dots by drawing only four straight lines, and never lifting your pen from the paper.



***If you want the answers to these questions or Degel to help you solve your real impossible problems, give us a call.***

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